

MINFENG, China

WMO: 518390

Lat: 37.067N

Lon: 82.717E

Elev: 1410

StdP: 85.50

Time Zone: 8.00 (E08)

Period: 82-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.9	-12.2	-21.9	0.6	-3.4	-20.6	0.7	-4.3	5.4	-2.0	4.5	-1.3	0.2	N/A	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	35.5	17.6	34.1	17.1	32.6	16.6	19.7	31.3	18.7	30.6	17.8	29.4	1.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.0	13.5	24.2	14.9	12.5	23.4	13.8	11.7	23.0	63.3	31.5	59.4	30.8	56.3	29.2	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	5.4	4.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-17.4	22.3	1.8	1.2	-18.7	23.1	-19.7	23.8	-20.8	24.4	-22.1	25.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	-4.8	-0.3	7.2	14.7	19.8	23.3	24.7	24.1	18.8	11.5	4.1	-3.0		
	DBStd	10.89	3.22	3.13	4.00	3.43	3.29	2.83	2.52	2.57	2.67	3.32	3.82	3.61		
	HDD10.0	1459	458	288	103	4	0	0	0	0	0	23	179	403		
	HDD18.3	3052	717	521	344	117	23	2	0	1	26	213	427	662		
	CDD10.0	2096	0	0	17	144	303	401	457	439	264	69	2	0		
	CDD18.3	647	0	0	0	8	67	152	199	181	39	1	0	0		
	CDH23.3	8067	0	0	1	219	879	1757	2303	2222	627	58	0	0		
	CDH26.7	3444	0	0	0	58	292	752	1092	1075	171	5	0	0		
(14) Wind	WSAvg	1.5	1.1	1.4	1.6	2.0	1.9	1.8	1.6	1.4	1.4	1.2	1.1	1.0		
(15) Precipitation	PrecAvg	51	2	1	2	3	5	15	10	6	6	0	0	1		
	PrecMax	141	8	7	10	27	25	65	36	56	43	3	2	4		
	PrecMin	17	0	0	0	0	0	0	0	0	0	0	0	0		
	PrecStd	28	2	2	3	6	6	15	9	11	11	1	0	1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.5	14.0	22.0	31.1	33.4	37.3	37.4	37.1	32.8	27.3	19.7	11.6		
		MCWB	0.4	3.4	7.6	12.3	14.5	16.8	17.8	18.6	16.6	11.7	7.1	2.4		
	2%	DB	5.2	10.6	20.1	28.1	31.7	34.3	35.7	35.2	30.2	25.1	16.3	7.4		
		MCWB	-1.3	2.1	6.8	11.4	14.1	16.7	17.5	17.9	14.9	10.7	5.5	0.5		
	5%	DB	3.2	8.6	18.3	25.8	29.9	32.5	34.1	33.8	28.7	22.6	14.0	5.0		
		MCWB	-2.3	0.8	6.2	10.2	13.4	16.4	17.5	17.2	14.1	9.6	4.4	-0.9		
	10%	DB	1.4	6.9	16.0	23.6	28.4	30.7	32.3	32.1	27.2	20.2	11.6	3.0		
		MCWB	-3.3	-0.2	5.0	9.1	12.9	15.5	17.1	16.9	13.3	8.6	3.2	-2.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.5	4.0	8.2	12.8	16.2	19.8	21.1	21.6	18.5	12.7	7.7	2.7		
		MCDB	7.8	13.2	20.7	30.4	29.4	32.3	31.2	32.7	29.5	24.1	17.9	10.8		
	2%	WB	-1.1	2.2	7.1	11.7	15.0	18.2	19.6	20.0	16.0	11.3	5.9	0.6		
		MCDB	4.6	9.9	19.4	26.9	28.6	30.6	31.0	31.7	27.6	23.7	15.3	7.0		
	5%	WB	-2.3	1.1	6.2	10.5	14.2	17.1	18.7	18.7	15.0	10.1	4.6	-0.8		
		MCDB	2.8	7.9	17.8	24.7	27.9	29.4	30.3	30.7	26.6	22.0	13.3	4.4		
	10%	WB	-3.3	0.0	5.2	9.4	13.4	16.2	18.0	17.7	14.1	8.9	3.5	-1.9		
		MCDB	1.2	6.4	15.4	22.5	26.6	28.6	29.4	29.3	25.2	19.4	11.3	2.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.8	12.1	13.6	14.3	13.3	12.8	12.8	14.4	15.8	16.6	14.0	11.4		
		MCDBR	14.3	15.0	16.8	19.6	17.5	16.8	17.3	18.6	18.3	20.1	17.4	14.4		
	5% WB	MCWBR	9.5	8.8	8.1	7.9	6.7	6.5	6.0	6.5	6.8	8.5	9.2	9.0		
		MCWBR	13.8	14.0	16.3	17.4	14.3	13.5	12.7	14.9	15.7	18.7	17.1	13.8		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.383	0.491	0.590	0.624	0.648	0.639	0.640	0.577	0.430	0.354	0.327		
	taud		2.146	1.969	1.702	1.528	1.515	1.525	1.569	1.556	1.631	1.878	2.065	2.124		
	Ebn at Noon		845	825	762	713	699	682	684	668	679	759	792	804		
	Edh at Noon		112	153	221	279	287	284	270	268	235	166	121	108		
(44) All-Sky Solar Radiation	RadAvg		2.92	3.79	5.19	6.14	6.79	6.67	6.44	6.02	5.46	4.72	3.42	2.74		
	RadStd		0.29	0.21	0.27	0.25	0.34	0.32	0.50	0.45	0.27	0.11	0.17	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page